

Smart Irrigation System Using IoT and Soil Moisture Monitoring

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Abstract—Water is one of the most important resources required for agriculture, and inefficient irrigation can result in considerable water wastage. Conventional irrigation methods often depend on manual observation and fixed watering schedules, which may result in over-irrigation or under-irrigation. This paper presents a Smart Irrigation System that uses soil moisture monitoring and Internet of Things (IoT) technology to automate the irrigation process. The proposed system uses a soil moisture sensor to continuously monitor the moisture level of the soil. An ESP32 microcontroller processes the sensor data and controls a water pump through a relay module. When the soil moisture level falls below a predefined threshold, the pump is automatically activated. When sufficient moisture is detected, the pump is switched off. Additional environmental parameters such as temperature and humidity can also be monitored. The system reduces manual intervention and can help improve water utilization. The proposed approach provides a simple, low-cost, and scalable solution for smart agriculture.

Index Terms—Smart Irrigation, Internet of Things, IoT, Soil Moisture, ESP32, Smart Agriculture, Water Management, Automation

I. INTRODUCTION

Agriculture is one of the major sectors that depends heavily on water availability. Traditional irrigation practices often involve manually supplying water to agricultural fields at fixed intervals. Such methods may lead to unnecessary water consumption because the actual moisture condition of the soil is not continuously considered.

The development of Internet of Things (IoT) technology has created opportunities for improving agricultural practices. IoT-based systems can collect information from sensors and process the information using microcontrollers. Based on the collected data, automated actions can be performed without continuous human intervention [1], [2].

A smart irrigation system can monitor soil moisture and automatically control water supply according to the moisture requirement of the soil. This approach can reduce unnecessary irrigation and improve water management [3], [4].

The proposed system uses an ESP32 microcontroller, soil moisture sensor, relay module, and water pump. The sensor provides soil moisture information to the controller. The controller compares the measured value with a predefined threshold and activates or deactivates the pump accordingly.

The major objectives of this research are:

- To monitor soil moisture continuously.
- To automate the irrigation process.
- To reduce unnecessary water usage.
- To minimize manual intervention.
- To develop a low-cost smart irrigation solution.
- To provide a system that can be extended with IoT monitoring capabilities.

II. LITERATURE REVIEW

Several researchers have investigated the use of IoT and sensor technologies in agriculture. IoT-based agricultural systems can collect environmental information and use the collected data for decision making [2], [5].

A sensor-based irrigation approach can use soil moisture information to determine whether irrigation is required. Previous research has demonstrated that automated irrigation can reduce unnecessary water application compared with fixed irrigation schedules [3], [4].

Wireless sensor networks have also been investigated for agricultural monitoring. Such systems allow environmental parameters to be collected from multiple locations in a field [6], [7].

Machine learning techniques have additionally been explored for irrigation prediction and crop monitoring. These methods can consider multiple parameters such as soil moisture, temperature, humidity, rainfall, and crop characteristics.

However, some existing systems may require expensive hardware, complex installation procedures, or continuous internet connectivity. The proposed system focuses on a relatively simple and low-cost architecture that can be expanded with additional sensors and IoT functionality.

III. PROPOSED SYSTEM

The proposed Smart Irrigation System automatically supplies water according to the moisture condition of the soil. The integration of sensors and IoT technologies provides a foundation for automated agricultural monitoring [2], [8].

The system consists of the following major components:

- 1) Soil moisture sensor
- 2) ESP32 microcontroller
- 3) Relay module
- 4) Water pump

- 5) Water tank
- 6) Temperature and humidity sensor
- 7) Power supply
- 8) IoT monitoring interface

The soil moisture sensor measures the moisture level of the soil. The sensor output is provided to the ESP32 controller. The controller compares the measured value with a predefined threshold.

If the soil is sufficiently dry, the ESP32 activates the relay, which turns ON the water pump. Water is supplied to the soil until the moisture reaches the required level. The controller then switches OFF the pump.

IV. SYSTEM ARCHITECTURE

The architecture of the proposed Smart Irrigation System is shown in Fig. 1. The architecture follows the general concept of connecting sensing devices, processing units, and communication systems for agricultural monitoring [1], [5].

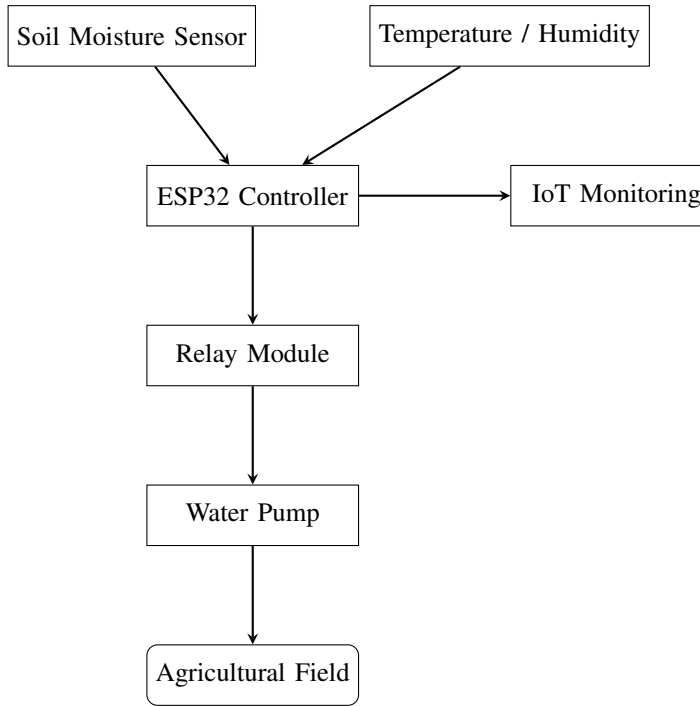


Fig. 1. Architecture of the Proposed Smart Irrigation System

V. HARDWARE REQUIREMENTS

The major hardware components required for the proposed system are listed in Table I. Sensor-based architectures are commonly used for collecting agricultural environmental information [6].

VI. SOFTWARE REQUIREMENTS

The software requirements for the system include:

- Arduino IDE
- Embedded C/C++
- ESP32 board support package

TABLE I
HARDWARE COMPONENTS

No.	Component	Purpose
1	Soil Moisture Sensor	Measures soil moisture
2	ESP32	Main controller
3	Relay Module	Controls the pump
4	Water Pump	Supplies water
5	DHT Sensor	Measures temperature and humidity
6	Water Tank	Stores water
7	Power Supply	Provides electrical power

- IoT monitoring platform
- Sensor libraries

The ESP32 is programmed using the Arduino IDE. The program reads the sensor values and makes irrigation decisions based on the predefined threshold.

VII. METHODOLOGY

The methodology consists of sensor data collection, processing, threshold-based decision making, and automated actuation. This type of sensor-driven architecture is consistent with approaches used in smart agriculture and automated irrigation systems [3], [4].

The proposed system operates through the following stages.

A. Soil Moisture Measurement

The soil moisture sensor measures the moisture condition of the soil. The sensor produces an electrical signal corresponding to the moisture level.

B. Data Processing

The ESP32 receives the sensor output and processes the measured value.

C. Threshold Comparison

The measured soil moisture is compared with a predefined threshold value.

D. Pump Control

If the measured moisture is below the threshold, the controller activates the water pump through the relay.

If the measured moisture reaches or exceeds the required level, the controller switches the pump OFF.

E. IoT Monitoring

The ESP32 can transmit sensor information to an IoT platform through Wi-Fi. This allows the user to monitor the irrigation system remotely. IoT platforms can provide communication and monitoring capabilities for agricultural applications [1], [8].

VIII. FLOWCHART

The overall operation of the proposed system is represented in Fig. 2.

IX. CONTROL ALGORITHM

The basic irrigation control algorithm is given in Algorithm 1.

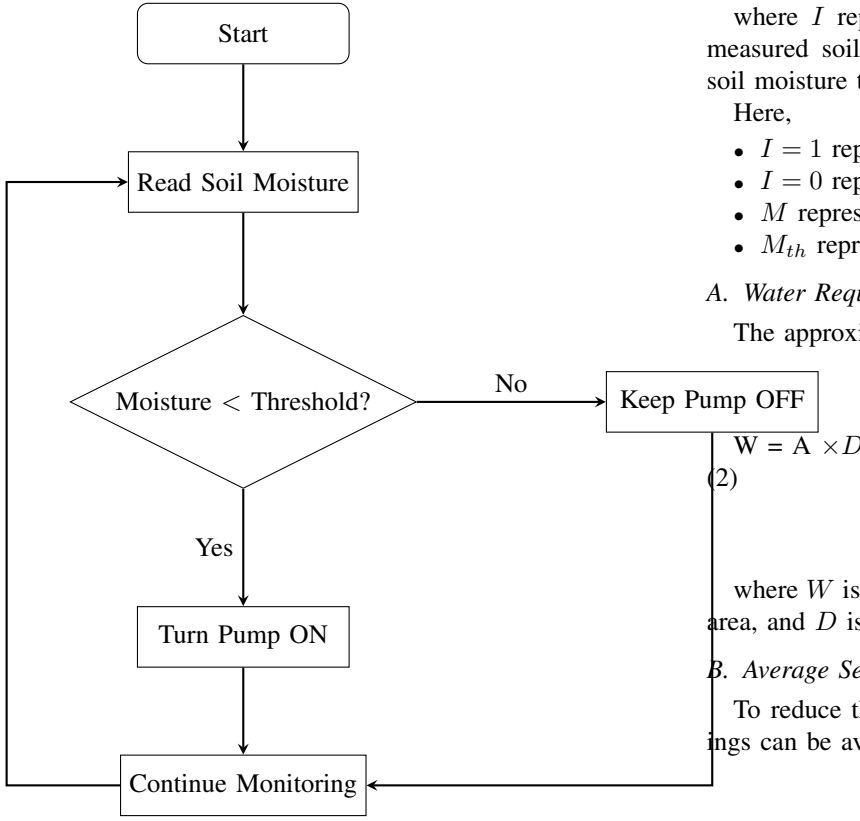


Fig. 2. Flowchart of the Smart Irrigation System

Algorithm 1 Smart Irrigation Control

- 1: Start
- 2: Initialize ESP32 and sensors
- 3: Read soil moisture value
- 4: Compare moisture with threshold
- 5: **if** moisture < threshold **then**
- 6: Turn ON water pump
- 7: **else**
- 8: Turn OFF water pump
- 9: **end if**
- 10: Send data to IoT platform
- 11: Repeat monitoring
- 12: Stop

X. MATHEMATICAL MODEL

The irrigation decision can be represented mathematically:

$$I = \begin{cases} 1, & M < M_{th} \\ 0, & M \geq M_{th} \end{cases} \quad (1)$$

where I represents the irrigation state, M represents the measured soil moisture, and M_{th} represents the predefined soil moisture threshold.

Here,

- $I = 1$ represents pump ON.
- $I = 0$ represents pump OFF.
- M represents measured soil moisture.
- M_{th} represents threshold moisture.

A. Water Requirement

The approximate water requirement can be represented as:

$$W = A \times D \quad (2)$$

where W is the required volume of water, A is the irrigated area, and D is the required water depth.

B. Average Sensor Reading

To reduce the effect of sensor noise, multiple sensor readings can be averaged:

$$M_{avg} = \frac{1}{n} \sum_{i=1}^n M_i \quad (3)$$

where M_i represents the individual sensor readings and n is the number of readings.

XI. SENSOR DATA REPRESENTATION

The sensor parameters can be represented as a feature vector:

$$X = [M \quad T \quad H \quad R] \quad (4)$$

where:

- M = Soil moisture
- T = Temperature
- H = Humidity
- R = Rainfall

Multiple sensor observations can be represented as a matrix:

$$X = \begin{bmatrix} M_1 & T_1 & H_1 & R_1 \\ M_2 & T_2 & H_2 & R_2 \\ M_3 & T_3 & H_3 & R_3 \end{bmatrix} \quad (5)$$

This representation allows multiple sensor readings to be stored and processed systematically.

XII. DATASET AND PARAMETERS

The proposed system can monitor multiple environmental parameters. An example parameter table is shown below.

TABLE II
INPUT PARAMETERS

Parameter	Symbol	Unit
Soil Moisture	M	%
Temperature	T	°C
Humidity	H	%
Rainfall	R	mm
Water Level	L	%

XIII. RESULTS AND DISCUSSION

The proposed system can be evaluated by observing the relationship between soil moisture and pump operation.

Table III provides an example of system observations.

TABLE III
SAMPLE IRRIGATION RESULTS

Soil Moisture	Pump Status	Irrigation
20%	ON	Required
30%	ON	Required
45%	OFF	Not Required
60%	OFF	Not Required
75%	OFF	Not Required

The results demonstrate the basic decision-making behavior of the system. When the soil moisture falls below the selected threshold, irrigation is activated. When sufficient moisture is available, the pump remains switched off.

The actual threshold should be selected according to the soil type, crop, environmental conditions, and irrigation requirements.

XIV. PROTOTYPE IMPLEMENTATION

A photograph of the developed hardware prototype can be included using the following code.

Replace `smart_irrigation.jpg` with the actual filename of your prototype image uploaded to Overleaf.

XV. PERFORMANCE COMPARISON

The proposed automated approach can be compared conceptually with conventional manual irrigation. Automated irrigation using sensor networks has previously been investigated as an alternative to conventional irrigation approaches [3], [4].

`smart_irrigation.jpg`

Fig. 3. Prototype of the Smart Irrigation System

TABLE IV
COMPARISON OF IRRIGATION METHODS

Parameter	Manual	Smart System
Automation	Low	High
Manual Effort	High	Low
Moisture Monitoring	Periodic	Continuous
Pump Control	Manual	Automatic
IoT Monitoring	No	Possible
Water Management	Variable	Data-based

XVI. ADVANTAGES

The major advantages of the proposed system are:

- Automatic irrigation based on soil moisture.
- Reduced manual intervention.
- Continuous monitoring of soil conditions.
- Potential reduction in unnecessary water usage.
- Low-cost hardware implementation.
- IoT connectivity can enable remote monitoring.
- The system can be expanded with additional sensors.

XVII. LIMITATIONS

The proposed system has certain limitations:

- Sensor accuracy can affect irrigation decisions.
- Soil moisture sensors require appropriate placement.
- The pump requires an adequate power supply.
- IoT monitoring depends on network availability.
- A single moisture sensor may not represent the moisture condition of a large agricultural field.

XVIII. FUTURE SCOPE

The proposed system can be further improved by integrating additional technologies.

Future developments may include:

- Weather forecasting integration.

- Rainfall prediction.
- Multiple soil moisture sensors.
- Mobile application monitoring.
- Cloud-based data storage.
- Machine learning based irrigation prediction.
- Solar-powered irrigation.
- Automated fertilizer management.
- Crop-specific irrigation recommendations.

Machine learning can also be introduced to predict irrigation requirements using historical soil, weather, and crop data. IoT and data analytics have been identified as important technologies for developing more advanced smart agriculture systems [2], [8].

XIX. CONCLUSION

This paper presented an IoT-based Smart Irrigation System for automated water management. The system uses soil moisture sensing to determine the irrigation requirement and an ESP32 microcontroller to control the water pump.

The proposed approach reduces the need for continuous manual monitoring and provides an automated mechanism for supplying water when the soil becomes dry. The system can also be extended with temperature, humidity, rainfall, and IoT monitoring capabilities.

The proposed architecture provides a simple foundation for developing smart agriculture applications. Further research can focus on larger agricultural fields, multiple sensors, weather prediction, machine learning, and solar-powered operation.

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